

Module 4: Responsible & Ethical AI

100 Tutorial Topics: Ethical Frameworks, Principles, Bias, Privacy, Governance, Regulation

GARP RAI Exam Preparation

T1: Responsible AI Definition

TOPIC: What is Responsible AI?

DEFINITION:

- Responsible AI refers to the ethical and responsible development, deployment, and use of artificial intelligence systems
- Aims to ensure that AI benefits society holistically while mitigating potential risks
- Involves aligning AI systems in a variety of ways to ensure they operate in a fair, transparent, and accountable manner

ALIGNMENT AREAS:

- **Ethical Standards:** Moral principles and values
- **Industry Standards:** Best practices and guidelines
- **Regulation:** Legal requirements and compliance
- **Legislation:** Statutory laws
- **Global Principles:** International norms and agreements

KEY GOAL:

- Benefit society holistically
- Mitigate potential risks
- Ensure fair, transparent, accountable operation

EXAM TRAP: Responsible AI = **ethical development and deployment** of AI systems!

T2: Purpose of Ethical Frameworks

TOPIC: Why Ethical Frameworks Matter in AI

PURPOSE:

- Help us decide how to design AI
- Help us deploy AI in a way that minimizes risks, harms, and wrongs
- Shape AI by helping define principles to abide by

KEY FUNCTIONS:

- 1 **Define Principles:** What to abide by
- 2 **Identify Biases:** Unacceptable biases
- 3 **Assess Transparency:** What kind is desirable
- 4 **Identify Stakeholders:** And their interests
- 5 **Promote Accountability:** Who is responsible
- 6 **Justify Decisions:** About design and implementation

KEY BENEFIT:

- Different ethical frameworks help make justifiable decisions
- Particularly important in pluralistic societies
- People hold diverse values and beliefs

EXAM TRAP: Ethical frameworks **guide decision-making** to minimize risks and harms!

T3: Practical Ethics Definition

TOPIC: What is Practical Ethics?

DEFINITION:

- The application of ethical theory and the development of good practices
- Goal: Solve real-world moral dilemmas
- Provide concrete guidance for moral decision making and problem solving

KEY ASPECTS:

1 Applied Focus:

- Addresses actual ethical dilemmas
- Domains: medicine, law, politics, business, AI
- Offers recommendations, not just theories

2 Multidisciplinary:

- Draws on moral philosophy
- Social sciences (psychology, sociology)
- Domain expertise, law, computer science

3 Context-Sensitive:

- Situational nuances matter
- No one-size-fits-all approaches

4 Pluralistic:

- Blends different ethical frameworks
- Aims for justifiable decisions

EXAM TRAP: Practical ethics = applying theory to real-world dilemmas!

TOPIC: Why Firms Should Consider Practical Ethics

KEY BENEFITS:

1 Building Trust and Reputation:

- Customers, employees, partners trust ethical companies
- Improves brand image
- Creates competitive advantage

2 Avoiding Scandals:

- Unethical behavior damages reputation and bottom line
- Ethics helps prevent major scandals

3 Attracting and Retaining Talent:

- Workers increasingly care about ethical practices
- Helps recruit and retain talent

4 Strengthening Culture:

- Nurtures teamwork, accountability, integrity
- Boosts morale and productivity

5 Supporting Risk Management:

- Reduces risk of safety failures, data breaches, non-compliance

EXAM TRAP: Ethics makes **good business sense** with multiple benefits!

T5: Ethics and Law Relationship

TOPIC: How Ethics and Law Relate

ETHICS AS COMPLEMENT TO LAW:

- Ethics helps ground, inform, and shape law
- Societies regulate behavior based on moral acceptability
- Ethics helps distinguish between just and unjust laws

LAWS ARE NARROW IN SCOPE:

- Establish minimal behavioral requirements
- Enable orderly interactions within basic fairness
- Provide a framework for society

ETHICS GOES BEYOND LAW:

- Identifies moral issues
- Reflects on the kind of society we want
- Makes recommendations for a good life
- More ambitious than law

EXAMPLE:

- It's not illegal to abandon friends, but it's immoral
- Laws provide minimums; ethics provides ideals

EXAM TRAP: Ethics **complements and informs** law - goes beyond minimum requirements!

TOPIC: Understanding Consequentialism

DEFINITION:

- An ethical theory that judges the morality of an action based on the consequences of that action
- Focuses on outcomes or results to determine whether an action is right or wrong

KEY PRINCIPLES:

1 Outcome Focus:

- What results from the action?
- Does it produce good or bad outcomes?
- Consequences determine moral value

2 Forward-Looking:

- Focus on future results
- Past actions judged by their effects
- Decisions based on expected outcomes

3 Circumstantially Relative:

- What's right depends on circumstances
- No absolute rules
- Context matters

EXAM TRAP: Consequentialism = judging actions by their consequences!

TOPIC: Utilitarianism - The Most Common Form of Consequentialism

CORE IDEA:

- Maximize overall utility
- Utility often defined as happiness, pleasure, or satisfaction of desires
- The morally correct action produces the greatest net utility for all affected

KEY THINKERS:

1 Jeremy Bentham:

- Founder of utilitarianism
- Hedonic calculus
- Focus on pleasure and pain

2 John Stuart Mill:

- Refined utilitarianism
- Quality of pleasures
- Higher and lower pleasures

APPLICATION TO AI:

- An AI system is morally good if it increases overall pleasure/happiness
- Actions are not intrinsically moral
- Derive moral value solely from results

EXAM TRAP: Utilitarianism = maximize overall utility/happiness!

T8: Consequentialism Advantages

TOPIC: Advantages of Consequentialism

KEY ADVANTAGES:

1 Single Quantifiable Metric:

- Provides one measure of moral value
- Utility can be calculated
- Clear decision-making framework

2 Impartiality:

- Everyone's utility counts equally
- No special treatment
- Objective calculation

3 Scientific Amenability:

- Utility can be measured
- Evidence-based decisions
- Quantifiable outcomes

4 Forward-Looking:

- Focuses on future outcomes
- Practical decision-making
- Real-world consequences

EXAM TRAP: Consequentialism provides a [single quantifiable metric](#) for moral decisions!

TOPIC: Criticisms of Consequentialism

KEY CHALLENGES:

1 Subjectivity of Utility:

- Choice of metric is subjective
- Different definitions of "good"
- Measurement difficulties

2 Unpredictable Consequences:

- Can't predict all outcomes
- Where does the calculation stop?
- Immediate vs. long-term consequences

3 Rights Violations:

- Can justify violating individual rights
- "The ends justify the means"
- Classic example: Killing one to save five

4 Doctor Example:

- Killing a patient to use organs saves five
- Maximizes utility but seems immoral
- Challenge for consequentialism

EXAM TRAP: Consequentialism can justify violating rights for the greater good!

TOPIC: Rule Consequentialism - A Modification

CORE IDEA:

- Judges acts by whether they adhere to utility-maximizing rules
- Not just by case-specific outcomes
- Addresses issues with act utilitarianism

HOW IT WORKS:

1 Follow General Rules:

- Rules that tend to maximize utility
- Rules against murder, lying, etc.
- Upheld even when breaking them may increase utility

2 Example - Doctor Case:

- Doctor should not kill patient
- Would undermine trust in medical system
- Long-term utility is maximized by following rules

3 AI Application:

- Should not violate rights even if efficient
- Long-term trust is valuable
- Rule-based protection

EXAM TRAP: Rule consequentialism = follow utility-maximizing rules!

TOPIC: Understanding Deontology

DEFINITION:

- An ethical framework that judges the morality of actions based on adherence to ethical duties and rules
- Rather than focusing on consequences
- Focus on moral obligations regardless of outcomes

KEY PRINCIPLES:

1 Duty-Based:

- Moral obligations are paramount
- We must do our duty
- Consequences don't override duties

2 Rule-Focused:

- Follow moral rules
- Universal principles
- Consistency across situations

3 Respect for Persons:

- People are ends in themselves
- Never treat as mere means
- Intrinsic value of individuals

EXAM TRAP: Deontology = judging actions by duties and rules!

TOPIC: Kant's Contribution to Deontology

KEY THINKER:

- Immanuel Kant is the most important proponent of deontology
- Emphasized moral principles, rationality, and duty
- Famous for not placing moral weight on consequences
- Believed lying is always wrong regardless of consequences

CORE CONCEPTS:

1 Categorical Imperative:

- Act only according to maxims that could be universal law
- "It's wrong to lie" - applicable universally
- No contradictions in universal application

2 Treating People as Ends:

- Never treat people as mere means to an end
- People have intrinsic value
- Respect their autonomy and dignity

3 Reason-Based Morality:

- Morality grounded in reason
- Not in emotions or consequences
- Rational beings can determine moral duties

EXAM TRAP: Kant = duty-based ethics focused on universal principles!

TOPIC: Understanding the Categorical Imperative

FIRST FORMULATION:

- "Act only according to that maxim whereby you can at the same time will that it should become a universal law"
- Can the rule be applied universally without contradiction?
- Example: Could "It's okay to lie" be a universal law?

SECOND FORMULATION:

- "Act in such a way that you treat humanity, whether in your own person or in the person of any other, always at the same time as an end and never merely as a means"
- Respect the dignity of all persons
- Don't use people as tools

APPLICATION TO AI:

- AI systems should respect human dignity
- People should not be treated as data points
- Transparency and consent are essential

EXAM TRAP: Categorical imperative = **act on rules that could be universal law!**

TOPIC: Contemporary Deontological Ethics

EVOLUTION OF DEONTOLOGY:

- Most contemporary deontologists care about consequences
- However, consequences are not the only moral consideration
- There are red lines that should not be crossed

KEY FEATURES:

1 Rights:

- Individuals have fundamental rights
- Rights cannot be violated even for good outcomes
- Examples: Privacy, autonomy, non-discrimination

2 Rules:

- Established ethical rules matter
- Professional codes of conduct
- Organizational policies

3 Principles:

- Ethical principles provide guidance
- Beyond consequences
- Deontological in nature

EXAM TRAP: Modern deontology considers consequences but has red lines!

TOPIC: Criticisms of Deontology

KEY CHALLENGES:

1 Rigidity:

- Can lead to rigid rule-following
- Even when consequences are disastrous
- Example: Kant said lying is always wrong

2 Conflicting Duties:

- What if duties conflict?
- Example: Duty to tell truth vs. duty to protect
- No clear resolution framework

3 Application Difficulty:

- Hard to apply in complex situations
- What rules apply?
- How to resolve conflicts?

AI APPLICATION:

- Can lead to overly rigid AI systems
- May not adapt to context
- Difficulty encoding all rules

EXAM TRAP: Deontology can be **too rigid** even when consequences are disastrous!

TOPIC: Understanding Virtue Ethics

DEFINITION:

- Emphasizes virtuous character traits and living a good life
- Rather than rules or consequences
- Has roots in ancient Greek philosophers like Aristotle

KEY PRINCIPLES:

1 Character Focus:

- What kind of person should I be?
- Cultivating virtues
- Becoming a good person

2 Virtues:

- Wisdom, courage, humanity
- Justice, temperance, generosity
- Character traits to cultivate

3 Practical Wisdom:

- Exercising judgment in situations
- Moderating emotions and appetites
- Appropriate behavior in context

EXAM TRAP: Virtue ethics = **focus on character** and **living a good life!**

TOPIC: Aristotle's Contribution to Virtue Ethics

HISTORICAL ROOTS:

- Virtue ethics has roots in ancient Greek philosophers
- Aristotle taught that happiness comes from a life guided by virtues
- Virtues are nurtured through practice and habit

KEY CONCEPTS:

1 Eudaimonia:

- Flourishing or living well
- Ultimate goal of human life
- Achieved through virtue

2 Virtues as Habits:

- Virtues are developed through practice
- Repeated actions create character
- Modeling virtuous exemplars

3 Golden Mean:

- Virtue lies between extremes
- Courage between cowardice and recklessness
- Balance in character

EXAM TRAP: Aristotle = virtues through practice for a good life!

T18: Key Virtues

TOPIC: Examples of Virtues

CORE VIRTUES:

1 Wisdom:

- Practical judgment
- Understanding what matters
- Sound decision-making

2 Courage:

- Facing challenges appropriately
- Moral courage
- Standing up for what's right

3 Humanity:

- Compassion and care
- Empathy for others
- Benevolence

4 Justice:

- Fairness and equity
- Giving proper due
- Social responsibility

5 Temperance:

- Self-control
- Moderation
- Balance

6 Generosity:

- Giving freely
- Sharing resources
- Beneficence

EXAM TRAP: Virtues include **wisdom, courage, justice, temperance, and generosity!**

TOPIC: Applying Virtue Ethics

PRACTICAL APPROACH:

1 Ask the Virtue Question:

- "What would a virtuous person do?"
- Consider role models and exemplars
- What would Jesus, Mohammed, Gandhi do?

2 Cultivate Virtues:

- Through practice and habit
- Learning from experience
- Modeling virtuous behavior

3 Consider Character:

- Not just discrete acts
- Life-long character development
- Holistic approach to morality

APPLICATION TO AI:

- Build AI that learns from experience like children
- Develop virtuous organizational culture
- Consider what responsible companies would do

EXAM TRAP: Virtue ethics asks "What would a virtuous person do?"!

TOPIC: Criticisms of Virtue Ethics

KEY CHALLENGES:

1 Lack of Clear Guidance:

- Less concrete than rules-based approaches
- Hard to know what virtue requires
- Situational judgment needed

2 Conflicting Virtues:

- Different virtues can conflict
- How to weigh courage vs. prudence?
- No clear resolution method

3 Cultural Relativism:

- Virtues vary across cultures
- What's virtuous in one context may not be in another
- Need for universal standards

VIRTUE ETHICIST RESPONSE:

- Practical wisdom helps navigate hard cases
- Other theories also have conflicts
- Consequences not always comparable

EXAM TRAP: Virtue ethics criticized for **lack of clear guidance!**

TOPIC: How Ethical Frameworks Complement Each Other

COMBINED APPROACH:

- Consequentialism, deontology, and virtue ethics are not mutually exclusive
- In practical ethics, they complement one another
- The best moral decision accords with all three

INTEGRATED DECISION-MAKING:

1 Consequentialist Analysis:

- What are the likely outcomes?
- Does it maximize good consequences?
- Consider all affected parties

2 Deontological Analysis:

- Does it respect rights and duties?
- Are there red lines being crossed?
- Is it in accordance with principles?

3 Virtue Ethics Analysis:

- What would a virtuous person do?
- Does it build character?
- Is it consistent with values?

EXAM TRAP: Frameworks are **complementary**, not mutually exclusive!

TOPIC: Medical Ethics - A Historical Overview

ANCIENT FOUNDATIONS:

- Hippocratic Oath (5th-3rd century BC)
- "Do no harm" to patients
- First known code of medical ethics

KEY DOCUMENTS:

① Nuremberg Code (1947):

- Response to Nazi medical experiments
- Informed consent requirement
- Research ethics foundation

② Declaration of Geneva (1948):

- Modern Hippocratic Oath
- International medical ethics

③ Declaration of Helsinki (1964):

- Research ethics guidelines
- Updated regularly
- Global standards

④ Belmont Report (1978):

- Respect for persons, beneficence, justice
- Research ethics framework
- US foundational document

EXAM TRAP: Medical ethics has long history and established frameworks!

TOPIC: Medical Ethics Development in the 1970s

FACTORS DRIVING EVOLUTION:

1 Concentration of Medical Care:

- Hospitals and depersonalized settings
- Loss of personal relationships
- Need for formal ethics

2 Rising Costs and Government Role:

- Increased government involvement
- Healthcare funding changes
- Resource allocation decisions

3 Patient Rights Movements:

- Civil rights outgrowth
- Demands for autonomy
- Informed consent

4 Medical Scandals:

- Tuskegee Syphilis Experiment
- Public outrage
- Call for reform

5 Rapid Technological Advances:

- Mechanical ventilators
- Organ transplantation
- New ethical dilemmas

EXAM TRAP: Multiple factors drove **development of medical ethics!**

TOPIC: What AI Ethics Can Learn from Medical Ethics

KEY LESSONS:

1 Structured Frameworks:

- Medical ethics has established principles
- AI needs similar foundations
- Clear ethical guidance

2 Professional Ethics:

- Doctors have ethical training
- AI professionals need ethics education
- Professional responsibility

3 Ethics Committees:

- Medical ethics committees
- AI needs similar oversight
- Multi-disciplinary review

4 Codes of Ethics:

- Medical codes are well-established
- AI codes are evolving
- Need for standards

EXAM TRAP: Medical ethics provides [model for AI ethics development!](#)

TOPIC: Five Core Principles of AI Ethics

THE FIVE PRINCIPLES:

1 Nonmaleficence:

- Obligation to avoid harming others
- "Do no harm"
- Prohibition of unjustified risks

2 Beneficence:

- Obligation to act for the benefit of others
- "Do good"
- Active help and support

3 Justice:

- Fairness, equality, impartiality
- Equitable distribution
- Non-discrimination

4 Autonomy:

- Respect for informed, un-coerced decisions
- Individual control
- Non-manipulation

5 Explainability:

- Transparent and understandable decisions
- Accountability
- Trust-building

EXAM TRAP: Five principles = nonmaleficence, beneficence, justice, autonomy, explainability!

TOPIC: The Principle of Nonmaleficence

CORE MEANING:

- Obligation to avoid harming others
- Prohibition on imposing unjustified risks
- "First, do no harm" (primum non nocere)

KEY ASPECTS:

1 Avoid Actual Harm:

- Don't directly harm others
- Example: Algorithms that discriminate
- Actions that cause injury

2 Avoid Unjustified Risk:

- Don't impose unnecessary risks
- Example: Untested algorithms
- Risks that outweigh benefits

3 Proportionality:

- Some risk is permissible
- If benefits justify risk
- If alternatives unavailable

EXAM TRAP: Nonmaleficence = "do no harm" - avoid harming others!

TOPIC: When is Risk Acceptable?

PRINCIPLES OF RISK ACCEPTANCE:

1 Benefit Justification:

- Risk is acceptable if benefits justify it
- Example: Clinical trials
- Risk is part of medical progress

2 No Alternatives:

- Risk acceptable if no alternatives
- Example: Emergency procedures
- Only option available

3 Who Bears the Risk:

- More acceptable if those at risk benefit
- Example: Clinical trial participants
- Unacceptable if others bear risk

AI APPLICATION:

- Imposing algorithmic risk on people who don't benefit
- Unacceptable to experiment on populations
- Especially if they don't gain

EXAM TRAP: Risk acceptable if **benefits justify it and risk-bearers benefit!**

TOPIC: The Principle of Beneficence

CORE MEANING:

- Moral obligation to act for the benefit of others
- Requires taking positive steps to help
- Moves beyond simply refraining from harm

KEY ASPECTS:

1 Active Help:

- Donating to charity
- Volunteering
- Providing resources and assistance

2 Beyond Nonmaleficence:

- Not just "do no harm"
- "Do good"
- Active beneficence

3 Deontological Duty:

- Duty-based obligation
- Not just about maximizing utility
- Required beyond consequences

EXAM TRAP: Beneficence = "do good" - actively help others!

TOPIC: Limits to the Duty of Beneficence

REASONABLE CONSTRAINTS:

1 Scarce Resources:

- Cannot help everyone
- Limited resources
- Prioritization needed

2 Competing Obligations:

- Multiple duties to balance
- Professional responsibilities
- Personal obligations

3 Reasonableness:

- There's only so much ethics can demand
- Demandingness constraint
- Proportionality matters

4 Recipient Autonomy:

- Unwanted "benefits" may be rejected
- Right to decide what's best
- Respect individual choices

EXAM TRAP: Beneficence has **reasonable limits** - cannot help everyone!

TOPIC: The Principle of Justice

CORE MEANING:

- Fairness, equality, impartiality, proportionality
- Giving each person their proper due
- Upholding duties toward fairness and equality

KEY APPLICATIONS:

1 Fair Treatment:

- No discrimination based on protected characteristics
- Equal opportunities
- Impartial algorithms

2 Equitable Distribution:

- Benefits and burdens fairly allocated
- Access to AI benefits
- Protection from AI harms

3 Accountability:

- Those responsible should be accountable
- Just processes
- Fair consequences

EXAM TRAP: Justice = fairness, equality, impartiality, and proportionality!

TOPIC: Four Types of Justice

FOUR TYPES:

1 Procedural Justice:

- Fair processes and impartiality
- Right structures in place
- Rule of law

2 Distributive Justice:

- Equitable allocation of benefits and burdens
- Fair distribution of resources
- Access to opportunities

3 Restorative Justice:

- Repairing harms
- Reconciling victims and offenders
- Making amends

4 Interactional Justice:

- Respect and fairness between individuals
- Interpersonal treatment
- Dignity and respect

EXAM TRAP: Four types of justice = procedural, distributive, restorative, interactional!

TOPIC: The Principle of Autonomy

CORE MEANING:

- Capacity to make informed, un-coerced decisions
- Respecting others' abilities to determine their own course
- Non-manipulation

KEY ASPECTS:

1 Informed Decisions:

- People need adequate information
- Transparency about options
- Understanding consequences

2 Un-coerced Decisions:

- No coercion or manipulation
- Free choice
- Voluntary consent

3 Capacity:

- Must have capacity to decide
- Exceptions: children, unconscious patients
- Surrogate decision makers when needed

EXAM TRAP: Autonomy = informed, un-coerced decisions!

TOPIC: Autonomy in Medical Context

INFORMED CONSENT:

- Patients have right to voluntary informed consent
- Right to refuse treatment
- Doctor's duty to inform
- Patient's right to decide

WHAT UNDERMINES AUTONOMY:

1 Coercion:

- Forcing decisions
- Threats or pressure
- Removing choice

2 Deception:

- Lying or misleading
- Withholding information
- Manipulating understanding

3 Undue Influence:

- Excessive persuasion
- Exploitation
- Imbalance of power

EXAM TRAP: Autonomy requires **informed consent** and **no coercion**!

TOPIC: Autonomy in the Context of AI

ETHICAL AI AND AUTONOMY:

- Ethical AI respects people's autonomy
- No coercive or manipulative tactics
- Technology should help people further their own goals
- Not serve third-party interests

RISKS TO AUTONOMY:

1 Manipulation:

- Recommendation algorithms
- Targeted content
- Behavioral influence

2 Coercion:

- Forced compliance
- No alternative
- System lock-in

3 Lack of Transparency:

- Users don't understand decisions
- Cannot make informed choices
- Autonomy undermined

EXAM TRAP: Ethical AI respects [autonomy](#) by [avoiding coercion and manipulation](#)!

TOPIC: The Principle of Explainability

CORE MEANING:

- AI systems should provide transparent explanations
- Understandable reasons for actions or decisions
- Also called "explicability"

WHY EXPLAINABILITY MATTERS:

1 Accountability:

- Hold designers and companies accountable
- Identify errors and biases
- Rectify unfair practices

2 Trust:

- Trust is fundamental for adoption
- Users trust understandable systems
- Transparency builds confidence

3 Ethical Compliance:

- Ensure adherence to ethical guidelines
- Respect individual rights
- Regulatory compliance

EXAM TRAP: Explainability = transparent and understandable explanations!

TOPIC: Levels and Types of Explainability

FOUR TYPES:

1 Local Explainability:

- Explains decisions on a single instance
- "Why was THIS decision made?"
- Example: Why was THIS loan denied?

2 Global Explainability:

- Explains overall model behavior
- "How does the model work generally?"
- Comprehensive understanding

3 Model-Specific Explainability:

- Techniques tailored to architecture
- Example: Decision tree rules
- Neural networks require different methods

4 Model-Agnostic Explainability:

- Works with any model
- Versatile and general
- LIME, SHAP are examples

EXAM TRAP: Explainability has **different levels and types** for different needs!

T37: Counterfactual Explanations

TOPIC: Counterfactual Explanations

WHAT THEY ARE:

- Hypothetical scenarios contrasting with actual decisions
- Show conditions for different outcomes
- Provide actionable insights

EXAMPLE - LOAN DENIAL:

- Actual: Loan denied
- Counterfactual: "You would have been approved if:
 - You had \$10,000 more in the bank
 - OR you earned \$1,000 more per month

BENEFITS:

- 1 **Transparency:**
 - Users understand decision factors
 - See what influences outcomes
- 2 **Actionable Insights:**
 - Know what to change
 - Path to different outcome
 - Empowerment
- 3 **Fairness:**
 - Detect biases
 - Understand disparate treatment
 - Identify improvement areas

EXAM TRAP: Counterfactual explanations show **what would lead to different outcomes!**

TOPIC: Understanding Bias in AI Systems

DEFINITION:

- Systemic deviation in output or impact
- Compared to a desired norm or standard
- Algorithm is biased when it deviates from intended function

EXAMPLE - HIRING ALGORITHM:

- Designed to identify best candidates
- But recommends based on race or sex
- Deviates from intended purpose
- This is problematic bias

KEY INSIGHT:

- No algorithm is perfectly objective
- Every algorithm reflects embedded values
- Values shape what's considered valuable

EXAM TRAP: AI bias = systemic deviation from intended function!

TOPIC: When is Bias Acceptable?

JUSTIFIABLE BIAS:

- Part of well-designed AI
- Optimizes for relevant criteria
- Example: Loan eligibility prioritizing bank balance
- Relevant to loan repayment

PROBLEMATIC BIAS:

- Results in unfairness
- Disadvantages for unjustifiable reasons
- Based on irrelevant characteristics
- Example: Race or sex discrimination

STATISTICALLY UNBIASED $\neq$ ETHICALLY ACCEPTABLE:

- Even unbiased algorithms can cause harm
- Example: Charging exorbitant fees to everyone
- Statistically fair but ethically problematic

EXAM TRAP: Not all biases are **problematic** - some are justifiable!

TOPIC: Four Primary Sources of Algorithmic Bias

1. Problem Specification:

- Goals contain inherent problems
- Target variables miss real-world objectives
- Example: Healthcare expenditure proxy

2. Data:

- Historical bias
- Sampling bias
- Selective labels problem

3. Modeling, Validation, and Design:

- Optimization choices
- Model selection
- Subgroup consideration

4. Deployment:

- Human deference
- Unintended effects
- Context changes

EXAM TRAP: Bias can enter at **any stage** of AI development!

TOPIC: Bias in Problem Specification

HOW IT OCCURS:

① Complex Goal Operationalization:

- Hard to capture real-world objectives
- Selected target variables may be poor proxies
- Example: Bank eliminating all risk

② Unintended Consequences:

- Bank lends only to affluent
- Excludes others
- Fairness issue

③ Proxy Problems:

- Healthcare expenditure proxy
- Favored richest patients, not sickest
- Systematically biased

KEY LESSON:

- Careful problem definition is essential
- Proxies must be thoughtfully chosen
- Goals should reflect true objectives

EXAM TRAP: Problem specification can introduce **bias** through **poor operationalization**!

TOPIC: Biases in Data

HISTORICAL BIAS:

- Data reflects past discrimination
- Example: Women excluded from banking
- Algorithm favors men
- Perpetuates historical patterns

SELECTIVE LABELS PROBLEM:

- Missing counterfactual outcomes
- No data on rejected applicants
- Can't know if better candidates existed
- Perpetuates existing patterns

SAMPLING BIAS:

- Non-random data sample
- Trends don't generalize
- Example: Data from young men only
- May not apply to women or elderly

EXAM TRAP: Data biases include **historical**, **selective labels**, and **sampling bias**!

TOPIC: Bias in Modeling, Validation, and Design

HOW BIAS EMERGES:

1 Optimization Functions:

- Choice of objective
- What is being optimized?
- May favor certain outcomes

2 Regression Models:

- Different models have different biases
- Choice matters
- Subgroup effects

3 Subgroup Consideration:

- How are subgroups handled?
- Are differences recognized?
- Fairness across groups

4 Information Presentation:

- How is data presented?
- Visual biases
- Interface effects

EXAM TRAP: Modeling choices can [introduce bias](#) even with good data!

TOPIC: Bias in Deployment

HOW IT OCCURS:

1 Human Deference:

- People defer to algorithm suggestions
- Even when limitations known
- Convenience factor

2 Reasons for Deference:

- Time-saving
- Perceived objectivity
- Self-doubt
- Responsibility shielding

3 Unintended Effects:

- Individuals relinquish responsibility
- Blame shifting
- Behavior changes

KEY LESSON:

- Oversight is essential
- Human judgment remains important
- Don't blindly trust automation

EXAM TRAP: Deployment can introduce **bias through human deference** to algorithms!

TOPIC: Why People Defer to Automation

CONVENIENCE:

- Time-saving
- Effort reduction
- Easy to follow

PERCEIVED OBJECTIVITY:

- Algorithms appear more objective
- People know their own fallibilities
- Creates self-doubt

RESPONSIBILITY SHIELDING:

- Deferring shares responsibility
- Going against algorithm exposes people
- Blame avoidance

CONSEQUENCES:

- Unintended effects
- Reduced oversight
- Bias perpetuation
- Accountability gaps

EXAM TRAP: People defer to algorithms for **convenience**, **objectivity perception**, and **responsibility shielding**!

TOPIC: When Bias Becomes Discrimination

LEGAL PERSPECTIVE:

- Depends on jurisdiction
- When it disfavors based on protected characteristics
- Race, sex, ethnicity, age
- Other classifications protected by law

PROTECTED CHARACTERISTICS:

- 1 Race
- 2 Sex/Gender
- 3 Ethnicity
- 4 Age
- 5 Religion
- 6 Disability
- 7 Sexual orientation

KEY POINT:

- Such disadvantages typically violate legal protections
- Proactive measures needed
- Continuous monitoring

EXAM TRAP: Discrimination = **disfavoring based on protected characteristics** in violation of law!

TOPIC: Understanding Fairness in AI

DEFINITION:

- Absence of bias or preference
- Based on irrelevant characteristics
- Algorithm is fair when it doesn't exhibit problematic biases

TWO TYPES:

① Group Fairness:

- Statistical criteria across groups
- Demographics mirror population
- Example: 51% female = 51% loan recipients

② Individual Fairness:

- Treating similar individuals similarly
- Challenge: Defining "similar"
- Focus on individual treatment

EXAM TRAP: Fairness = [absence of problematic biases!](#)

TOPIC: Why Fairness Can't Be Automated

MATHEMATICAL IMPOSSIBILITY:

- When base rates are unequal
- Cannot satisfy all fairness measures simultaneously
- Demographic parity, predictive rate parity, equal opportunity
- Trade-offs are unavoidable

EXAMPLE - CRIMINAL RISK ASSESSMENT:

- Men statistically more likely to commit certain crimes
- Algorithm assigns higher risk to man
- Man argues unfair treatment
- Woman argues unfair if assigned equal score

KEY POINT:

- Fairness is a moral judgment
- Not purely mathematical
- Compromises and trade-offs needed

EXAM TRAP: Fairness cannot be automated when **base rates are unequal!**

TOPIC: Procedural Fairness in AI

DEFINITION:

- Fair processes and impartial procedures
- Not just fair outcomes
- Right structures in place

JUSTICE SYSTEM ANALOGY:

- Procedural fairness: rule of law
- Outcome fairness: appropriate punishments
- Mistakes can be justifiable with fair processes
- Ways to right wrongs

AI APPLICATION:

- Ethics committees
- Established procedures
- Appeal processes
- Accountability structures

EXAM TRAP: Procedural fairness = fair processes and impartial procedures!

TOPIC: Strategies to Avoid Problematic Biases

KEY APPROACHES:

1 Technological Solutions:

- Fairness toolkits (Aequitas, AIF360)
- Internal auditing systems
- Bias reports from providers
- But technology alone is insufficient

2 Data Quality:

- Diverse, updated, accurate data
- Representative samples
- Free from past discriminatory tendencies

3 Auditing:

- Algorithmic auditing is best approach
- Private companies offer services
- Independent assessment

4 Ethical Committees:

- Discuss biases
- Consider trade-offs
- Ensure procedural fairness

EXAM TRAP: Bias mitigation requires **multiple approaches beyond technology!**

TOPIC: Understanding Privacy in AI

DEFINITION:

- Privacy to extent others don't have access to personal information
- In reference to some person or institution
- In reference to some personal data point

WHY PRIVACY MATTERS:

1 Protection from Abuse:

- More knowledge enables interference
- Power imbalances
- Control over individuals

2 Autonomy:

- Privacy supports self-determination
- Freedom from surveillance
- Personal space

3 Dignity:

- Respect for persons
- Not being reduced to data
- Human worth

EXAM TRAP: Privacy = lack of access to personal information!

TOPIC: Understanding Data Security

DEFINITION:

- Doing what is necessary to prevent unauthorized access
- Protecting data from attacks and theft
- Ensuring data integrity

KEY COMPONENTS:

1 Physical Security:

- Hardware protection
- Storage device security
- Physical access controls

2 Administrative Controls:

- Access permissions
- Organizational policies
- Employee training

3 Software Applications:

- Encryption
- Authentication
- Monitoring systems

EXAM TRAP: Data security = preventing unauthorized access to data!

TOPIC: Why Privacy is an Ethical Issue

WRONGS:

- Violates moral and legal rights
- Treats people instrumentally
- Reduces people to data points
- Respect for autonomy

HARMS TO INDIVIDUALS:

- Discrimination
- Blackmail
- Exposure and shaming
- Identity theft

HARMS TO INSTITUTIONS:

- Lawsuits and liability
- Fines and penalties
- Reputational damage

HARMS TO SOCIETY:

- National security risks
- Democratic threats
- Blackmail of officials

EXAM TRAP: Privacy violations cause **wrongs and harms** to individuals, institutions, and society!

TOPIC: Personal Data - A Toxic Asset

WHY "TOXIC"?

① Cheap to Mine:

- Easy to collect
- Low cost
- Widespread availability

② Valuable:

- Profitable for companies
- Marketing and targeting
- Business insights

③ Sensitive:

- Privacy concerns
- Vulnerable to abuse
- High risk

④ Hard to Keep Safe:

- Security challenges
- Attacker advantage
- Constant threats

KEY INSIGHT:

- Defenders are at disadvantage
- Attacker chooses time and method
- With enough resources, breach is inevitable

EXAM TRAP: Personal data is a **toxic asset** - valuable but risky!

T55: Data Minimization

TOPIC: The Principle of Data Minimization

CORE IDEA:

- Collect only necessary data
- Fulfill specific purpose
- Most effective privacy protection

WHY IT MATTERS:

1 Risk Reduction:

- Less data = less risk
- Fewer targets
- Reduced exposure

2 Respect for Persons:

- Privacy rights
- Avoid unnecessary intrusion
- Proportionality

3 Regulatory Compliance:

- GDPR requirement
- Privacy laws
- Best practice

KEY POINT:

- Only collect data when benefit outweighs disadvantages
- Data subjects deserve protection
- Minimal data collection is respectful

EXAM TRAP: Data minimization = **collect only necessary data!**

TOPIC: The Right to be Forgotten

DEFINITION:

- Person's right to have private information removed
- From Internet searches and other directories
- When data is inadequate, irrelevant, or excessive

WHEN IT APPLIES:

1 Incorrect Information:

- Data that is wrong
- Inaccurate records
- False information

2 Irrelevant Information:

- No longer relevant
- Outdated
- No legitimate purpose

3 Excessive Information:

- More than needed
- Proportionality
- Over-collection

EXAM TRAP: Right to be forgotten = removal of private information when inadequate or irrelevant!

TOPIC: Giving People Control Over Their Data

KEY RIGHTS:

1 Consent Management:

- Easy to deny consent
- Easy to withdraw consent
- Clear options

2 Access:

- See personal data
- See inferences made
- Understand data use

3 Transfer:

- Data portability
- Move data to other services
- Freedom to choose

4 Deletion:

- Right to delete data
- Remove personal information
- Erasure

EXAM TRAP: Control over data = [consent](#), [access](#), [transfer](#), and [deletion rights](#)!

TOPIC: Contextual Integrity in Privacy

CORE IDEA:

- Data use should adhere to contextual norms
- When data is transferred to different context, norms are violated
- Context matters

EXAMPLES:

1 Banking Context:

- Data given for transaction
- Should not be sold to marketing company
- Violates expectations

2 Healthcare Context:

- Patient data given to doctor
- Should not end up with data brokers
- Violates trust

3 Social Media Context:

- Expectations about data use
- Context-specific norms
- Violations cause backlash

EXAM TRAP: Contextual integrity = data use should respect context-specific norms!

TOPIC: Best Practices for Data Deletion

WHY DELETE DATA:

1 Protect Individuals:

- Reduce exposure
- Privacy protection
- Security

2 Maintain Accuracy:

- Personal data changes quickly
- People change jobs, houses, preferences
- Outdated data is inaccurate

3 Reduce Liability:

- Less data = less risk
- Fewer targets
- Smaller breach impact

BEST PRACTICE:

- Regular deletion schedules
- Expiry dates for data
- Not collected with intention of keeping forever
- Routine data deletion

EXAM TRAP: Delete data **as soon as no longer necessary** - routine practice!

TOPIC: Summary of Privacy Principles

KEY PRINCIPLES:

- ➊ **Right to Privacy:**
 - Moral and legal right
 - Universal Declaration of Human Rights
 - Constitutional protections
- ➋ **Data Minimization:**
 - Collect only necessary data
 - Most effective protection
- ➌ **Right to be Forgotten:**
 - Remove inadequate data
 - Irrelevant or excessive
- ➍ **Control Over Data:**
 - Consent, access, transfer, deletion
- ➎ **Contextual Integrity:**
 - Respect context-specific norms
- ➏ **Data Deletion:**
 - Delete when no longer necessary
- ➐ **Data Security:**
 - Encryption, anonymization, differential privacy

EXAM TRAP: Multiple principles guide **ethical privacy practices!**

TOPIC: Why AI is Difficult to Govern

KEY CHALLENGES:

1 Power Asymmetries:

- Tech companies more powerful than governments
- Wealth, influence, expertise
- Regulation difficulties

2 Lack of Transparency:

- Private company development
- Limited public information
- Safety testing unknown

3 Lack of Interpretability:

- Neural networks are "black boxes"
- Proxy variables obscure
- Hard to audit

4 Lack of Ethics Structures:

- No required ethics education
- No licensing
- No oversight committees

EXAM TRAP: AI governance faces **multiple interconnected challenges!**

TOPIC: Power Asymmetries in AI Governance

THE PROBLEM:

- Tech companies often more powerful than governments
- Wealth: Massive resources
- Influence: Lobbying and public relations
- Expertise: Advanced technical knowledge

CONSEQUENCES:

1 Regulation Challenges:

- Governments struggle to keep up
- Companies resist regulation
- Asymmetric negotiation

2 Information Asymmetry:

- Companies know more
- Governments lack technical expertise
- Hard to regulate effectively

3 Accountability Gaps:

- Who holds companies accountable?
- Regulatory capture risk
- Limited enforcement

EXAM TRAP: Tech companies may be **more powerful than some governments!**

T63: Lack of Transparency

TOPIC: Transparency Challenges in AI

WHY IT'S A PROBLEM:

- Most AI systems developed by private companies
- Not subject to transparency requirements
- Limited information about:
 - Training datasets
 - Testing and tuning
 - Safety measures

CONSEQUENCES:

1 No Public Oversight:

- Academics can't study systems
- Journalists can't investigate
- Public is in the dark

2 Regulatory Challenges:

- Regulators lack information
- Can't assess risks
- Enforcement difficulties

3 Trust Issues:

- Lack of transparency erodes trust
- Suspicion about practices
- Public concern

EXAM TRAP: Private companies are **not** subject to same transparency requirements!

TOPIC: Interpretability Challenges

THE BLACK-BOX PROBLEM:

- Neural networks are "black boxes"
- Even experts can't fully understand
- Backpropagation doesn't provide insight
- Hard to trace decisions

WHY IT MATTERS:

1 Accountability:

- Can't hold accountable what you can't understand
- Responsibility gaps
- Who is responsible?

2 Auditing:

- Hard to audit black boxes
- New methods needed
- Limited effectiveness

3 Governance:

- Can't govern what you can't understand
- Regulatory challenges
- Oversight difficulties

EXAM TRAP: Neural networks are **black boxes** - even experts can't fully understand them!

T65: Proxy Variables Challenge

TOPIC: Proxy Variables and Governance

WHAT ARE PROXIES?

- Techniques to simplify training
- Represent complex objectives with simpler metrics
- Example: Word count as proxy for article quality

THE CHALLENGE:

1 Alignment Issues:

- Proxy may not align with true objective
- Optimization for wrong goal
- Unintended consequences

2 Opacity:

- If proxies are unknown
- Hard to understand model behavior
- Governance difficulties

3 Scale and Complexity:

- Billions of parameters
- Hard to trace processes
- Audit challenges

EXAM TRAP: Unknown proxies make governance difficult!

TOPIC: Lack of AI Ethics Structures

COMPARISON WITH MEDICAL ETHICS:

Medical Ethics	AI Ethics
Bioethicists	Few AI ethicists
Ethical codes	Developing codes
Ethics committees	Rare committees
Required coursework	Optional courses
Licensed professionals	No licensing
International codes	Fragmented guidance

KEY GAP:

- Computer scientists can graduate without ethics training
- No certification or licensing
- Boards rarely include AI ethicists
- No oversight committees

EXAM TRAP: AI ethics lacks the **established structures** of medical ethics!

TOPIC: The Regulatory Gap

CURRENT STATE:

- Few national laws (China exception)
- No specialized agencies
- No government-mandated auditing
- Voluntary frameworks only

HISTORICAL PARALLEL:

- FDA established for food and drugs
- Other regulatory agencies developed
- AI may follow similar path
- EU leading with AI Office

WHY REGULATION MATTERS:

- AI is international technology
- Cross-border data flows
- Need for cooperation
- International standards needed

EXAM TRAP: Few national AI laws exist (except China) - no specialized agencies!

TOPIC: Unpredictability in AI Systems

EMERGENT BEHAVIOR:

- Properties not explicitly coded
- Capabilities that surprise
- Mistakes that are unexpected
- Interactions that are unforeseen

CONSEQUENCES:

1 Safety Risks:

- Unintended outcomes
- Harmful behavior
- System failures

2 Governance Challenges:

- Hard to regulate unpredictable systems
- Misuse difficult to prevent
- Unknown risks

3 Trust Issues:

- Surprises undermine trust
- Unpredictable = unreliable
- Adoption barriers

EXAM TRAP: Emergent behavior = **unexpected capabilities not explicitly coded!**

TOPIC: LLMs and Truth-Tracking

HOW LLMS WORK:

- Statistical inferences and probabilistic guesses
- Designed for plausible responses
- Not based on truth or knowledge

RISKS:

1 Misinformation:

- Plausible but false information
- Hard to detect
- Spread at scale

2 Physical Safety:

- Incorrect medical advice
- Dangerous recommendations
- Real-world harm

3 Legal Risks:

- Libel and defamation
- Copyright issues
- Speech-related risks

EXAM TRAP: LLMs give **plausible, not necessarily truthful** responses!

TOPIC: Privacy and Copyright Challenges

PRIVACY CONCERNS:

- Data scraped from internet
- Personal data of unsuspecting users
- GDPR compliance uncertain
- Right to access, modify, delete unclear

COPYRIGHT CONCERNS:

- LLMs ingest copyrighted material
- Books, articles, images
- Lawsuits in process
- Legal uncertainty

GOVERNANCE NIGHTMARE:

- Machine-created speech
- Company ownership
- Data acquisition unclear
- Regulatory challenges

EXAM TRAP: Scraping data raises **privacy and copyright concerns!**

TOPIC: Global AI Regulatory Landscape

KEY PLAYERS:

1 European Union:

- GDPR (Privacy)
- DMA (Digital Markets)
- DSA (Digital Services)
- AI Act (Comprehensive AI Law)

2 United States:

- No federal privacy law
- State-level privacy laws
- AI Executive Order
- NIST AI RMF

3 China:

- Generative AI Measures
- PIPL (Privacy)
- Algorithmic Recommendation Rules

EXAM TRAP: Different regions have **different** regulatory approaches!

TOPIC: European GDPR

KEY FACTS:

- Implemented May 2018
- Regulates personal data
- Applies to data controllers and processors
- Extraterritorial jurisdiction

DATA SUBJECT RIGHTS:

- 1 Receive concise information
- 2 Access personal data
- 3 Request erasure
- 4 Object to processing

OBLIGATIONS:

- Notify breaches within 72 hours
- "Legitimate interest" required
- Significant fines for non-compliance

EXAM TRAP: GDPR = [European data protection regulation](#) with global reach!

TOPIC: Digital Markets Act (DMA)

KEY FACTS:

- Passed by European Union in 2022
- Aims for fair and competitive digital marketplace
- Targets "gatekeeper" platforms

KEY PROVISIONS:

1 Ban on Self-Preferencing:

- Can't favor own services
- No ranking own products higher
- No making switching difficult

2 Open Access to Data:

- Provide data to third parties
- Enable innovative services
- Foster competition

3 Transparency Obligations:

- Transparent about algorithms
- Allow understanding of operations

4 No Tying and Bundling:

- Can't require unwanted services
- Core services accessible separately

EXAM TRAP: DMA targets **gatekeeper platforms** for fair competition!

TOPIC: Digital Services Act (DSA)

KEY FACTS:

- Adopted October 2022
- Came into force 2023
- Applies to online intermediaries and platforms
- 19 companies currently beholden

KEY OBLIGATIONS:

1 Prevent Illegal Content:

- Proactively identify and remove
- Hate speech, CSAM, counterfeit goods
- Clear reporting mechanisms

2 Transparency:

- Publish content-moderation policies
- Provide access to moderation data

3 Address Disinformation:

- Labels on political advertising
- Promote fact-checking

4 Protect from Algorithmic Bias:

- Ensure algorithms not biased
- Transparent about algorithms

5 Opt-Out of Personalized Content:

- Very large platforms must allow
- No tracking for recommendations

EXAM TRAP: DSA addresses illegal content, transparency, and disinformation!

TOPIC: The EU AI Act

KEY FACTS:

- Approved March 2024
- Council approved May 2024
- Establishes common regulatory framework
- Risk-based approach

RISK CATEGORIES:

1 Unacceptable Risk (Banned):

- Biometric categorization using sensitive characteristics
- Untargeted facial image scraping
- Emotion recognition in workplace
- Social scoring
- Subliminal manipulation

2 High-Risk:

- Impact assessments required
- Thorough records
- Oversight measures

3 Limited/Minimal Risk:

- Transparency obligations
- Labeling requirements

EXAM TRAP: EU AI Act = [first comprehensive AI regulation](#) - risk-based approach!

TOPIC: EU AI Act - Specific Requirements

TRANSPARENCY REQUIREMENTS:

- Notify people when interacting with chatbot
- Notify when using biometric systems
- Notify when using emotion recognition
- Label deepfakes and AI-generated content
- Design systems to make AI-generated media detectable

HIGH-RISK REQUIREMENTS:

- Impact assessment on rights
- Keep thorough records of:
 - Datasets used
 - Programming and training methodologies
 - Oversight measures

PENALTIES:

- €35 million or 7% of global turnover
- €7.5 million or 1.5% of turnover
- Depends on offense and company size

EXAM TRAP: EU AI Act has **significant penalties** for non-compliance!

T77: US Privacy Regulations

TOPIC: US Privacy Regulations

NO FEDERAL PRIVACY LAW:

- United States lacks comprehensive federal privacy law
- Patchwork of state-level regulations
- HIPAA for health information

KEY STATE LAWS:

1 California Consumer Privacy Act (CCPA):

- Most comprehensive state privacy law
- Right to know, access, delete
- Right to opt out of sale
- Right to non-discrimination

2 Virginia Consumer Data Protection Act (CDPA):

- Modeled after CCPA
- Additional protections for sensitive data
- Genetic and biometric data

3 Colorado Privacy Act (CPA):

- Right to opt out of targeted advertising
- Data minimization provisions

4 Illinois BIPA:

- Biometric Information Privacy Act
- Notification, access, deletion rights
- Civil remedies for violations

EXAM TRAP: US has **patchwork of state privacy laws** - no federal law!

TOPIC: US AI Regulatory Framework

KEY INITIATIVES:

1 Executive Order on AI (2023):

- Safe, secure, trustworthy development
- Address AI-enabled harms
- International cooperation
- Safety testing requirements

2 NIST AI RMF:

- Voluntary framework
- Govern, map, measure, manage
- Flexible and adaptable
- Non-prescriptive

3 AI Safety and Security Board:

- Department of Homeland Security
- Critical infrastructure focus
- Recommendations and guidance

EXAM TRAP: US has **executive orders and voluntary frameworks** - no federal AI law yet!

TOPIC: China AI Regulatory Framework

KEY REGULATIONS:

① Generative AI Measures (2023):

- Applies to generative AI services
- Prohibits illegal content
- Prevents discriminatory content
- Protects privacy and personal information

② Personal Information Protection Law (PIPL):

- Comprehensive privacy law
- Data minimization requirements
- User consent requirements

③ Algorithmic Recommendation Rules:

- Transparency in algorithm decision-making
- User consent requirements
- Data minimization

EXAM TRAP: China has [comprehensive AI regulations](#) - among first globally!

TOPIC: Comparing Global AI Regulation

EUROPEAN UNION:

- Comprehensive, risk-based approach
- GDPR for privacy
- AI Act for AI specifically
- Strong enforcement
- Extraterritorial reach

UNITED STATES:

- No federal AI law
- State-level privacy laws
- Executive orders and frameworks
- Voluntary compliance
- Sector-specific approach

CHINA:

- Comprehensive AI regulations
- Generative AI measures
- Strong state control
- Rapid development

EXAM TRAP: EU leads with [comprehensive regulation](#); US relies on [voluntary frameworks](#)!

T81: Ethics as Complement to Law

TOPIC: How Ethics Complements Law

ETHICS GROUNDS LAW:

- Ethics helps ground, inform, and shape law
- Societies regulate behavior based on moral acceptability
- Ethics helps distinguish between just and unjust laws

LAWS ARE MINIMAL:

- Establish minimal behavioral requirements
- Enable orderly interactions
- Provide framework for society

ETHICS GOES BEYOND:

- Identifies moral issues
- Reflects on kind of society we want
- Makes recommendations for a good life
- More ambitious than law

EXAMPLE:

- Law: Not illegal to abandon friends
- Ethics: Abandoning friends is immoral
- Laws provide minimums; ethics provides ideals

EXAM TRAP: Ethics **complements and informs** law - goes beyond minimum requirements!

TOPIC: Ethics in AI Regulation

RELATIONSHIP:

- AI ethics gaining importance
- Some feel laws also needed
- Ethics informs regulatory approaches

EXAMPLES:

- GDPR: Not designed for AI but relevant
- EU AI Act: Based on ethical principles
- NIST AI RMF: Incorporates ethical considerations

KEY INSIGHT:

- Regulations increasingly require ethical considerations
- Ethics frameworks underlay responsible AI approaches
- Ethical principles guide regulatory development

EXAM TRAP: Ethics **informs and shapes** AI regulation!

TOPIC: Comprehensive Responsible AI Framework

KEY COMPONENTS:

1 Ethical Foundations:

- Consequentialism
- Deontology
- Virtue ethics

2 Core Principles:

- Nonmaleficence, beneficence, justice, autonomy, explainability

3 Bias Mitigation:

- Identify sources
- Implement strategies
- Continuous monitoring

4 Privacy Protection:

- Data minimization
- Security measures
- User control

5 Governance:

- Ethics committees
- Auditing
- Accountability

EXAM TRAP: Responsible AI requires **comprehensive approach** across all areas!

TOPIC: The Role of Ethics Committees in AI

PURPOSE:

- Discuss possible problematic biases
- Make decisions about trade-offs
- Consider multiple ethical frameworks
- Ensure procedural fairness

BENEFITS:

1 Diverse Perspectives:

- Different viewpoints
- Multiple expertise areas
- Balanced considerations

2 Structured Decision-Making:

- Systematic review
- Transparent processes
- Documented reasoning

3 Accountability:

- Clear responsibility
- Audit trail
- Defensible decisions

EXAM TRAP: Ethics committees help **discuss and address biases** through diverse perspectives!

TOPIC: Continuous Monitoring and Evaluation

IMPORTANCE:

- First strategy to prevent reputational loss
- Identify risks proactively
- Address them proactively

HOW TO IMPLEMENT:

1 Organizational Governance:

- Mechanisms for regular monitoring
- Clear responsibilities
- Reporting structures

2 Risk Assessment:

- Regular evaluations
- Identify new risks
- Update mitigation strategies

3 Performance Tracking:

- Monitor AI system performance
- Track fairness metrics
- Detect drift or degradation

EXAM TRAP: Continuous monitoring is **essential** for responsible AI!

TOPIC: Stakeholder Engagement and Communication

DEFINITION:

- Regularly engage with stakeholders
- Customers, employees, regulators
- About AI initiatives and their impacts

BENEFITS:

- Clear communication prevents misunderstandings
- Builds trust
- Early identification of concerns
- Collaborative problem-solving

APPROACH:

- Regular updates
- Transparent about challenges
- Responsive to feedback
- Inclusive of diverse perspectives

EXAM TRAP: Stakeholder engagement **builds trust and prevents misunderstandings!**

TOPIC: AI-Related Crisis Management

DEFINITION:

- Perform risk analysis
- Inform development of crisis-management plan
- Tailored to AI-related incidents

COMPONENTS:

- Communication strategies
- Potential steps to rectify issue
- Clear lines of responsibility
- Escalation procedures

BENEFITS:

- Faster response to incidents
- Minimize reputational damage
- Demonstrate competence and responsibility
- Rebuild trust more effectively

EXAM TRAP: Crisis management plan = **proactive preparation!**

TOPIC: Training for Board and C-Suite

WHY LEADERSHIP EDUCATION MATTERS:

- Wide range of risks from AI
- Financial, legal, reputational consequences
- Responsible AI is a strategic imperative
- Leadership drives organizational culture

KEY TOPICS FOR TRAINING:

1 AI Risks:

- Bias and discrimination
- Privacy violations
- Regulatory non-compliance
- Reputational damage

2 Ethical/Responsible AI Practices:

- Fairness principles
- Governance structures
- Risk mitigation

3 Strategic Considerations:

- Competitive advantage
- Trust building
- Long-term value

EXAM TRAP: Leadership must be [educated about AI risks and responsible practices!](#)

TOPIC: Summary - Bias, Discrimination, and Fairness

KEY CONCEPTS:

1 Bias:

- Systemic deviation from norm
- Can be justifiable or problematic
- Sources: Problem, data, modeling, deployment

2 Discrimination:

- Bias that disfavors protected characteristics
- Legally prohibited
- Requires proactive measures

3 Fairness:

- Absence of problematic bias
- Group and individual fairness
- Cannot be fully automated

EXAM TRAP: Bias management requires **continuous effort across the AI lifecycle!**

TOPIC: Tools for Bias Mitigation

AVAILABLE TOOLKITS:

- 1 **Aequitas:**
 - Tests models for different bias metrics
 - Across different groups
 - Auditing tool
- 2 **AI Fairness 360 (AIF360):**
 - IBM's fairness toolkit
 - Benchmark for testing
 - Multiple fairness metrics
- 3 **Internal Auditing Systems:**
 - Companies can create own systems
 - Identify potential biases
 - Continuous monitoring

EXTERNAL OPTIONS:

- Third-party auditing services
- Independent assessors
- Bias reports from providers
- Fresh perspective

EXAM TRAP: Multiple tools available for **identifying and mitigating bias!**

T91: Data Quality Importance

TOPIC: Data Quality and Bias Prevention

KEY DATA QUALITY FACTORS:

1 Diversity:

- Include all relevant groups
- Avoid homogeneity
- Represent all populations

2 Accuracy:

- Correct and reliable data
- Avoid errors
- Verify sources

3 Representativeness:

- Reflect target population
- Avoid sampling bias
- Generalizable results

4 Currency:

- Up-to-date data
- Reflect current reality
- Avoid outdated patterns

SYNTHETIC DATA:

- Fabricated datasets
- No privacy risks
- Can be designed to avoid biases
- May be less precise

EXAM TRAP: Data quality helps [avoid biases](#) through diversity and representativeness!

TOPIC: The Importance of Algorithmic Auditing

WHY AUDITING MATTERS:

- Likely the best way to identify biases
- Fresh and diverse perspective
- Independent assessment
- Continuous improvement

WHAT AUDITING INCLUDES:

1 Technological Tools:

- Statistical analyses
- Fairness metrics
- Bias detection

2 Human Review:

- Diverse perspectives
- Contextual understanding
- Qualitative assessment

3 Documentation:

- Findings and recommendations
- Remediation plans
- Continuous monitoring

EXAM TRAP: Algorithmic auditing is **best way to identify and correct biases!**

TOPIC: Summary of Privacy Principles

KEY PRINCIPLES:

1 Right to Privacy:

- Moral and legal right
- Universal Declaration of Human Rights

2 Data Minimization:

- Collect only necessary data

3 Right to be Forgotten:

- Remove inadequate data

4 Control Over Data:

- Consent, access, transfer, deletion

5 Contextual Integrity:

- Respect context-specific norms

6 Data Deletion:

- Delete when no longer necessary

7 Data Security:

- Encryption, anonymization, differential privacy

EXAM TRAP: Multiple principles guide **ethical privacy practices!**

TOPIC: Summary of Ethical Frameworks

CONSEQUENTIALISM:

- Focus: Outcomes and consequences
- Key Question: "What maximizes good outcomes?"
- Strength: Single quantifiable metric
- Weakness: Can justify rights violations

DEONTOLOGY:

- Focus: Duties and rules
- Key Question: "What are our moral obligations?"
- Strength: Respects rights and dignity
- Weakness: Can be rigid

VIRTUE ETHICS:

- Focus: Character and virtues
- Key Question: "What kind of person should I be?"
- Strength: Holistic and aspirational
- Weakness: Lack of clear guidance

EXAM TRAP: Each framework offers [valuable insights](#); best decisions integrate all three!

TOPIC: Summary of AI Ethics Principles

FIVE CORE PRINCIPLES:

1 Nonmaleficence:

- "Do no harm"
- Avoid unjustified risks
- Proportional risk-benefit

2 Beneficence:

- "Do good"
- Active help
- Advance welfare

3 Justice:

- Fairness and equality
- Impartiality
- Non-discrimination

4 Autonomy:

- Informed decisions
- Non-coercion
- Respect for persons

5 Explainability:

- Transparency
- Understandable decisions
- Accountability

EXAM TRAP: Five principles provide **comprehensive ethical framework** for AI!

TOPIC: Summary of AI Governance Challenges

KEY CHALLENGES:

- ❶ **Power Asymmetries:**
 - Tech companies more powerful than governments
- ❷ **Lack of Transparency:**
 - Private development, limited information
- ❸ **Lack of Interpretability:**
 - Black-box models, proxy variables
- ❹ **Lack of Ethics Structures:**
 - No required education, licensing, committees
- ❺ **Lack of Regulation:**
 - Few national laws, no specialized agencies
- ❻ **Unpredictability:**
 - Emergent behavior, unexpected capabilities
- ❼ **Truth-Tracking:**
 - LLMs give plausible, not truthful, responses

EXAM TRAP: AI governance faces **multiple interconnected challenges!**

TOPIC: Summary of Global Regulatory Landscape

EUROPEAN UNION:

- GDPR: Comprehensive privacy regulation
- DMA: Fair competition in digital markets
- DSA: Illegal content, transparency, disinformation
- AI Act: Risk-based AI regulation
- European AI Office: Enforcement body

UNITED STATES:

- No federal privacy law
- State-level privacy laws (CCPA, CDPA, CPA, BIPA)
- Executive Order on AI
- NIST AI RMF (voluntary)
- AI Safety and Security Board

CHINA:

- Generative AI Measures
- PIPL (privacy)
- Algorithmic Recommendation Rules

EXAM TRAP: Different regions have [different regulatory approaches](#) - EU most comprehensive!

TOPIC: Summary of Practical Ethics

KEY ASPECTS:

- Applied focus on actual ethical dilemmas
- Multidisciplinary approach
- Context-sensitive emphasis on situational nuances
- Pluralistic blend of different frameworks

BUSINESS BENEFITS:

- Building trust and reputation
- Avoiding scandals
- Attracting and retaining talent
- Strengthening culture
- Supporting risk management
- Guiding innovation
- Promoting long-term thinking

KEY INSIGHT:

- Practical ethics makes good business sense
- Benefits overlap and reinforce each other
- Ethics frameworks underlay responsible AI

EXAM TRAP: Practical ethics = [applying theory to real-world dilemmas](#) with multiple benefits!

TOPIC: Implementing Responsible AI

KEY STEPS:

1 Establish Ethical Foundations:

- Adopt ethical frameworks
- Define core principles
- Create ethics committees

2 Implement Bias Mitigation:

- Identify sources
- Use appropriate tools
- Regular auditing

3 Protect Privacy:

- Data minimization
- Strong security
- User control

4 Ensure Explainability:

- Transparent decisions
- Understandable explanations
- Accountability

5 Establish Governance:

- Oversight structures
- Continuous monitoring
- Stakeholder engagement

EXAM TRAP: Responsible AI requires **comprehensive implementation** across all areas!

TOPIC: Comprehensive Summary - Responsible & Ethical AI

ETHICAL FOUNDATIONS:

- Consequentialism (outcomes and utility)
- Deontology (duties and rules)
- Virtue Ethics (character and virtues)

CORE PRINCIPLES:

- Nonmaleficence, beneficence, justice, autonomy, explainability

KEY PRACTICES:

- Bias identification and mitigation
- Privacy protection and data minimization
- Transparency and explainability
- Human oversight and accountability

GOVERNANCE AND REGULATION:

- Ethical committees and structures
- Algorithmic auditing
- Compliance with regulations (GDPR, EU AI Act)
- Continuous monitoring and improvement

EXAM SUCCESS: Understand **ethical frameworks, principles, governance, and regulation** for responsible AI!